

# 3rd Grade Balloon Racer

(Coach Scheduled Event - 45 Minutes)



**Description:** Teams will design, build, and race an air-powered balloon device to travel a distance along a fishing line track in the shortest amount of time. They will answer questions on Newton's Laws of Motion and air pressure.

## **Participants per assigned Team Number: 2**

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

### **Safety Equipment - Teams Must Bring:**

- **Safety glasses labeled ANSI Z87+ (impact rated)**
- **All competitors must wear their eye protection during the entire event.**
- **If a team does not have the required eye protection, they will be given the opportunity to obtain it, time allowing, but will not receive extra time.**
- **If a team is unable to obtain eye protection, the team will not compete and will receive a no-show score.**

Impact Safety Glasses Required



**Eyeglasses are not safety rated!**

### **Teams must also bring:**

- Writing Utensils
- One pair of scissors

### **Materials Provided at the Event:**

- 3- 30 cm (12 in) Balloons
- 1 - Paper Smoothie straw - 10 mm (0.4 in) diameter, 23 cm (9 in) length (wide opening)
- 2 - Paper straws - 6 mm (0.24 in) diameter, 20 cm (7.75 in) length
- 3 - #16 Rubber band
- 1 - 22 cm by 30 cm (8 ½ in x 11 in) 60 to 90 pound cardstock
- 50 cm (20 inches) of 2.54 cm (1 in) masking tape
- Competition Track and a similar Practice Track
- Balloon inflater

### **The Competition:**

**The Written Assessment:** The team members will take a written assessment on their understanding of Newton's Laws of Motion and air pressure. The test will take place simultaneously during the construction of the team racer.

- As a part of the written test the team will estimate the time needed for their racer to travel the competition day distance.

### **Construction Phase (30 minutes):**

- Teams may cut their materials to any size they deem appropriate,
  - Teams may use as much or as little of their materials to build their racer, except that the

team's design must have one straw to be taped to the outside of their balloon to use for the Competition Track line to pass through. That straw may be cut, but must still accommodate the fishing line track for the vehicle to run on the track.

- Teams may request one additional balloon without penalty if they break one, but requests for any additional materials beyond that places the team in Tier 3. This means that the team may still participate but will be placed below all teams that do not violate this rule.
- Teams may have 1 untimed practice run on a practice track that is roughly the same length as the competition track to test their racer within the 30 minutes build time with permission of the Event Supervisor
  - Releasing a racer down either the practice track or the competition track without permission of the Event Supervisor during both the testing and scoring phases of the competition will result in a team being placed in Tier 3.
  - Practice runs will not occur after the 30-minute build time

### ***The Competition Track:***

- The Competition Track will be a fishing line along which the balloon racer travels. It will have one end of the fishing line attached permanently on one side of the room. The other end will be clipped to something at the same height as the permanent end. The track will be high enough off the floor so that the racer will not touch the ground.
- The length of the competition Track will be released the day of the competition and is subject to space limitations. It will be no more than 6 meters and no less than 3 meters.

### ***Official Testing:***

- The team will feed the non-permanent end of the competition track through one straw attached to their racer to prepare it for the run. The Event Supervisor will then clip the non-permanent end so that the track is parallel to the ground.
- Each team will have one officially scored run
- The team must state that they are ready to release the racer and must wait for the Event Supervisor to acknowledge they are ready to time the run.
- After the Event Supervisor's acknowledgement, a team member will make a loud announcement of, "3, 2, 1, LAUNCH!" and release the racer.
- Team members may not push their racer during the release. Teams that push their racer at any time will be placed in Tier 3, which means that they will be scored behind all other teams.
- The Event Supervisor will record the time for the racer from the start until the finish point to the precision of their timing device.
- If the racer does not reach the finish line, the Event Supervisor will record how close to the finish line the front of the racer is. That racer will be placed in Tier 2 and the recorded distance will substitute for the Time Score.

### **Scoring:**

**Final Score (FS) = Assessment Score + Time Score, max 100 points**

- Assessment score =  $(TAS \div CAS) \times 50$ ; rounded to the 100th place value; maximum 50 points
  - Team Assessment Score (TAS) = team's score on the assessment
  - Competition Assessment Score (CAS) = the highest assessment score for all teams
- Time Score =  $(CT \div TT) \times 50$ ; rounded to the 100th place value; maximum 50 points
  - Competition Time (CT) = the lowest Time for all teams
  - Team Time (TT) = the team's Time
- Tiers
  - Tier 1 - the racer is not in Tier 2 or Tier 3. Tier 1 teams score before Tier 2 or Tier 3 no matter their score.

- Tier 2 - the team's racer does not make it to the finish line
- Tier 3 - the team requests additional build materials beyond a second balloon, team takes a disallowed practice run, or team pushes their racer during their official test.

### **Scoring Example**

- Competition Assessment Score (CAS) = 15. The team scores 9 (TAS) on the assessment.  
Assessment Score for the team =  $(9 \div 15) \times 50 = 30$  points
- Competition Time (CT) = 1.45 seconds. The team's time (TT) is 3.56 seconds.  
Time Score for the team for the team =  $(1.45 \div 3.56) \times 50 = 20.36$  points
- Final Score (FS) =  $30 + 20.36 = 50.36$  points

### **Tiebreakers:**

- 1st - The team with the closest estimate to their travel time
- 2nd - The better score on the written test
- 3rd - Select questions on the test

### **Resources:**

- [NASA - What Is Aerodynamics?](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

# Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



## Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

### CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

### CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles